



UnionPay Integrated Circuit Card Specifications — Product Specifications

Part X Enhanced Low-Value Payment Specification



Version 2014

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Summary of Revisions

The change listed below is associated with the current version.

Description of Change	Where to look
Revised: Updated “Enhanced low-value payment products suggestion: 1” to “Enhanced stored-value mode low-value payment products: 1; Enhanced credit mode low-value payment products: 0”	6.2 Table 1

1 Application Scope

This book applies to all UPI participants.

2 Normative References

These specifications reference the provisions of the following documents, and these provisions may therefore be considered to be provisions of the specifications themselves. For all dated reference documents, all later amendments (except errata corrections) or modifications are not necessarily applicable to these specifications. For this reason it is recommended that all parties associated with this agreement conduct all necessary research and come to an understanding regarding what newer versions of these documents remain applicable. For all non-dated documents, their most recent version may be considered to be applicable to these specifications.

UnionPay Integrated Circuit Card Specifications--Basic Specifications Part 1: Debit/Credit Application Specifications

UnionPay Integrated Circuit Card Specifications--Basic Specifications Part 2: Debit/Credit Application Specifications

UnionPay Integrated Circuit Card Specifications--Basic Specifications Part 3: Debit/Credit Application Terminal Specifications

UnionPay Integrated Circuit Card Specifications--Basic Specifications Part 4: Debit/Credit Application Security Specifications

UnionPay Integrated Circuit Card Specifications--Basic Specifications Part 5: Contactless Type IC Card Payment Specifications

UnionPay Integrated Circuit Card Specifications--Basic Specifications Part 6: Low-Value Payment Specifications Based on Debit/Credit Applications

UnionPay Integrated Circuit Card Specifications--Product Specifications Part 2: Contactless Type IC Card Low-Value payment Expansion Application Specifications

UnionPay Integrated Circuit Card Specifications--Product Specifications Part 6: Dual Currency Low-Value Payment Specifications Based on Debit/Credit Applications

3 Terms and definitions

The following terms and definitions are applicable to *UnionPay Integrated Circuit Card Specifications*.

3.1 Enhanced Low-Value Payment

In response to market demands, upgrades and supplements on UICS low-value contactless payment products. Within this specification, the meanings of UnionPay enhanced low-value payment and enhanced low-value payment are the same.

3.2 Abnormal E-Cash Money Deducted Phenomenon

A rare and abnormal transaction phenomenon where a terminal denies a contactless E-cash transaction after the funds have already been successfully withdrawn from a card.

3.3 Dormant Card

An enhanced low-value payment card that is placed into a dormant status after its dormant card date hasn't been updated for at least 365 natural days.

3.4 Hot Card

A list of card numbers stored within a terminal; the terminal will refuse these cards if an offline transaction attempt is made.

3.5 Enhanced fDDA

The fast DDA algorithm utilized by enhanced low-value payment applications.

3.6 Command

A message sent from the terminal to IC cards to initiate one operation or request a response.

3.7 Cryptogram

Results of encrypted operation.

3.8 Electronic Cash (EC)

Function supporting micro payment based on debit/ credit application.

3.9 Electronic Cash Balance

A counter that displays the amount on given card that is available for offline spending.

3.10 Function

A sequence combined by one or more commands, which complete the whole or part of transaction.

3.11 Load

A process increases the electronic cash balance in the card. The loading may be carried out through multiple ways, such as transferring cash from the master account or depositing cash or transferring funds from other accounts. However, the electronic cash balance after loading shall not exceed the electronic cash balance limit.

3.12 Unload

The process of decreasing an electronic cash balance until it reaches 0. The result of unloading is the complete removal of an electronic cash balance and have it either transferred to the cardholder's main account or returned to the cardholder as cash.

3.13 Response

A message returned to the terminal after the IC card has completed the task required by the command message.

3.14 Script

A command or command sequence sent to a terminal from an issuer with the intent of inputting that command into an IC card.

3.15 Transaction Log

A record of recent transaction details that allows for an understanding of transaction history; this includes contactless electronic cash logs, load/unload transaction logs, etc.

3.16 Terminal

Devices installed at transaction locations that are used to connect to IC cards and complete financial transactions. They include device interface which may allow for interfacing with other component such as main computer systems, etc.

4 Symbols and Abbreviations

fDDA	Fast DDA
FCI	File Control Information
ARQC	Authorization Request Cryptogram
ARPC	Authorization Response Cryptogram
ATC	Application Transaction Counter
ATM	Automated Teller Machine
EC	Electronic Cash
EMV	Europay MasterCard VISA
CDOL	Card Risk Management Data Object List
CAM	Card Authentication Method
CVM	Cardholder Verification Method
IACs	Issuer Action Code
PIN	Personal Identification Number
TC	Transaction Certificate

5 Overall Introduction

The primary functions of UnionPay enhanced low-value payment products are the following:

- All UICS standard debit/credit/E-cash applications;
- Enhanced low-value payment cards issued in non-mainland China shall comply with require *UICS Product Specification – Part 6*; Enhanced low-value payment cards issued domestically may comply with *UICS Product Specification – Part 6*;
- The newly added functions below:
 - Dormant card management
 - Mutual security authentication mechanisms
 - Sensitive information management
- Special card settings: During card personalization, E-cash terminal indicator (Tag 9F7A) is not set into PDOL in order to disenable the contact E-cash function.

6 Transaction Flow

6.1 Contact Transaction Flow

The same as UICS standards; please see the relevant contents of *UICS Product Specification – Part 6*.

6.2 Contactless Transaction Flow

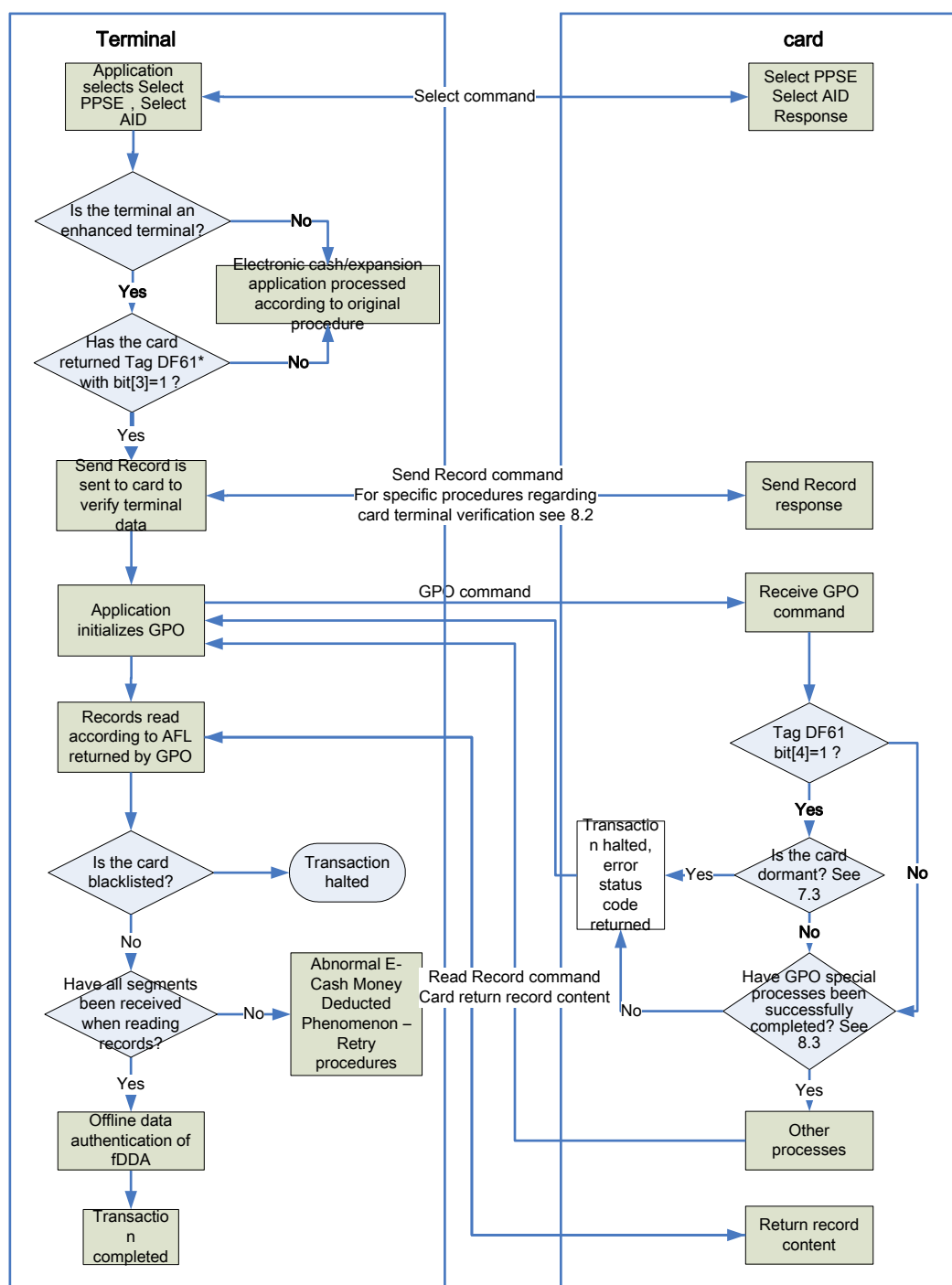


Figure 1 Enhanced Low-Value Payment Product Contactless Transaction Flow

Note:

——*See Table 1 for the description of Tag DF61 and Appendix A for corresponding personalization and security requirements.

——Card verification of terminal legality is conducted during the GPO special processing stage; see Section 8.3.

——Hot card's transaction should be rejected after the card number has been read by the terminal but before the reading of the final record has been completed.

Table 1 Card Add-Value Function Indicator (Tag "DF61")

Value/ Type		Description	Length (in bytes)	Existence
'DF61'		Card add-value function indicator	2	M
'var'		Card add-value function indicator	1	M
b	[8]	For extended applications based on Contactless low-value payment use. Defined in <i>UICS Product Specification – Part 2</i>	1	M
	[7-5]	RFU ('000')		
	[4]	1: Card supports dormant card management 0: Card does not support dormant card management Enhanced low-value payment products suggestion: 1		
	[3]	1: Card supports mutual security authentication 0: Card does not support mutual security authentication Enhanced stored-value mode low-value payment products: 1 Enhanced credit mode low-value payment products: 0		
	[2]	For extended applications based on		

		Contactless low-value payment use. Defined in <i>UICS Product Specification – Part 2</i>		
	[1]	For extended applications based on Contactless low-value payment use. Defined in <i>UICS Product Specification – Part 2</i>		
		RFU	var	O

7 Dormant Card Management

7.1 Overview

Dormant cards refer to enhanced low-value payment cards that have been placed into a dormant status after their transaction date stamp hasn't been updated for at least 365 natural days. When an enhanced low-value payment card begins an offline electronic cash transaction it will first verify that the transaction date stamp is earlier than the current date fewer 365 days. If it is, the card is in a dormant state; it will refuse the transaction and respond with "9408", transaction failed. For cards already in a dormant state, only by conducting an online transaction will an issuer decide whether or not approve the transaction. If the online transaction is approved, then after the transaction is successfully completed, the transaction date stamp will be updated automatically.

7.2 Newly Added Internal Data Element Description

Transaction date stamp, a new internal data element, is stored within the card and record the date of the last successfully completed online debit/credit transaction.

This internal data element is initialized during the personalization process and set the same value as the application effective date (Tag 5F25).

If the following two requirements are satisfied, the card will update the transaction date stamp and complete the second GAC command response in one atomic operation.

——Issuer authentication has been initiated and completed successfully, whether the issuer authentication indicator bit is mandatory or not ;

——Issuer request has been approved and the card has approved the transaction in the 2nd GAC.

The implementation method and security requirements of the internal data element "transaction dates tamp" is the same as those internal data elements described in *UICS Basic Specification – Part 2*, such as "Issuer Authentication Failure Indicator".

7.3 Transaction Process - Special Procedures

During the GPO processing, if the card requires dormant card management, then the transaction properties must be determined firstly. If the terminal requires online processing, then the transaction date stamp determination step will be skipped and transaction processes will continue; if it is an offline transaction, then decide whether the transaction date stamp is at least 365 days older than the current date. If it is, then the card is in a dormant state and the transaction will be refused; if not, then the transaction processes will continue.

7.4 Combined with Hot Card Mechanism

When an enhanced low-value payment card conducts an E-cash transaction with a terminal, during the GPO phase it will be determined whether or not the card is dormant; if it is, then the transaction will be refused. Terminal blacklists do not contain dormant cards, and for this reason they only store blacklists for one year. This effectively reduces the number of hot card numbers for the terminal, but to control risks the issuer must make a record of all hot card, including hot cards that have already gone dormant. In real applications, dormant card management reduces terminal burden and increases retrieval efficiency.

8 Mutual Security Authentication Mechanism

8.1 Overview

So as to increase the security and dependability of contactless E-Cash transactions and protect the interests of cardholders, the card and terminal must both authenticate the other. This mutual security authentication mechanism is only applicable for contactless enhanced low-value payment transactions.

Offline data authentication is the primary method by which enhanced low-value payment applications guarantee the legality of cards and terminals. Divided by the verification objects, there are two types of authentication:

——Legality authentication of cards by terminals

——Legality authentication of terminals by cards

The authentication of cards by terminals is the same as current UICS requirement; please see Chapter 5 of *UICS Basic Specification - Part 4*.

Cards utilize RSA key algorithms to verify the legality of terminals using the following steps:

- a) Once the UICS application has been selected, Tag DF61 is returned by card and used to determine whether or not the card is an enhanced low-value payment card; if not, then the original E-Cash process is executed;
- b) The terminal uses a Send Record command to send the signature algorithm identifier, CA public key index, terminal public key indicator, terminal public key remainder, and the terminal public key certificate to the card; the card sends the card random number to the terminal in return;
- c) The terminal uses the terminal private key to sign the transaction date, card random number, and terminal random number; the Send Record command then transmits these signatures to the card;
- d) The terminal transmits GPO instructions; the card executes the process described in Section 8.3 and determines whether or not the process requires terminal signature verification using the "Card Additional Processing: (Tag "9F68") security mechanism.

8.2 Diagram

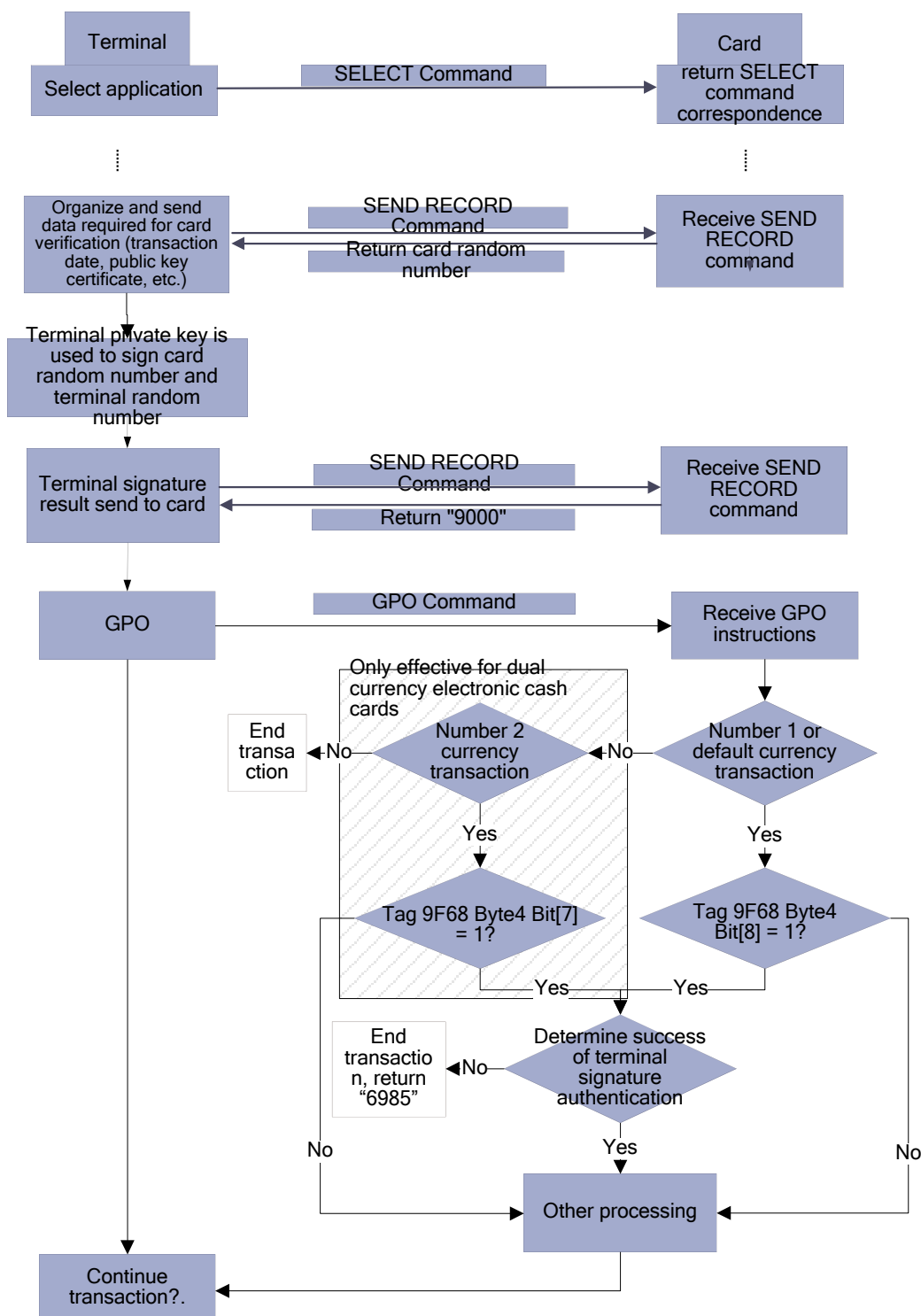


Figure 2 Card-Terminal Authentication Method Process Diagram

8.3 GPO Command Processing – Special Explanation

Once the card has received the GPO command it immediately initiates the following inspections:

—If the card is a dual currency electronic cash card then the transaction currency code is used to determine if the transaction will be conducted in currency one or currency two; if it is a single currency electronic cash transaction then the default currency will be used;

—When the transaction currency is either the first or default currency, the card will inspect the 8th bit of byte-4 of “Card Additional Processing” (Tag “9F68”) and determine whether or not Mutual authentication is necessary for that currency. If authentication is compulsory, then the signature sent from the terminal will be authenticated; if the authentication is successful then the transaction will proceed; if not, the card will stop the transaction and return a “6985” error code. If authentication is not required, the transaction will proceed normally;

—When the transaction currency is the second currency, the card will inspect the 7th bit of byte-4 of “Card Additional Processing” (Tag “9F68”) and determine whether or not mutual authentication is necessary for that currency. If authentication is compulsory, then the signature sent from the terminal will be authenticated; if the authentication is successful then the transaction will proceed; if not, the card will stop the transaction and return a “6985” error code. If authentication is not required, the transaction will proceed normally;

8.4 Data Element Description

Enhanced low-value payment product specifications expand upon “Card Additional Processing” (Tag “9F68”); see details in the table below.

Table 2 Card Additional Processing (Tag “9F68”)

Byte	Bit	Explanation
1	8	1 – Supports low-value inspection 0 – Does not support low-value inspection
	7	1 – Supports low-value and CTTA inspection 0 – Does not support low-value and CTTA inspection
	6	1 – Supports low-value or CTTA inspection 0 – Does not support low-value or CTTA inspection
	5	1 – Supports new card inspection 0 – Does not support new card inspection
	4	1 – Supports excessive PIN resubmission inspection 0 – Does not support excessive PIN resubmission inspection

	3	1 – Allows non-matching currency transactions 0 – Does not allow non-matching currency transactions
	2	1 – Card prefers to select contact-type debit/credit connections 0 – Card does not select contact-type debit/credit connections
	1	1 – Returns available offline spending amount 0 – Does not return available offline spending amount
2	8	1 – Supports prepay 0 – Does not support prepay
	7	1 – Does not allow non-matching currency transactions 0 – Allows non-matching currency transactions
	6	1- If the card is new and the terminal only offers offline support, then the transaction will be refused 0- If the card is new and the terminal only offers offline support, the transaction will not be refused
	5	1- qUICS offline approval of transaction; card records transaction log 0- qUICS offline approval of transaction; card does not record transaction log
	4-1	RFU
3	8	1 – Matching currency transaction supports online PIN 0 – Matching currency transaction does not support online PIN
	7	1 – Non-matching currency transaction supports online PIN 0 – Non-matching currency transaction does not support online PIN
	6	1 – Card requires CVM for non-matching currency transactions 0 – Card does not require CVM for non-matching currency transactions

	5	1 – Supports signature 0 – Does not support signatures
	4-1	RFU
4	8	1 – Mutual authentication for first/default currency is mandatory 0 – Mutual authentication for first/default currency is optional
	7	1 – Mutual authentication for second currency is mandatory 0 – Mutual authentication for second currency is optional
	6-1	RFU

8.5 Terminal Public/Private Key Management and Signature Calculations

8.5.1 Terminal Public Key Retrieval

Terminal public key retrieval uses the following procedures:

- 1) During the GPO instruction special processing phase, the CA public key and its associated algorithm is used to apply a specified restoration function (for details about restoration functions, please see Chapter 9 of *UICS Basic Specification – Part 4*) to the terminal public key certificate. If the end of the restored data is not equal to “BC”, then the card's verification of the legality of the terminal will fail.

Table 3 Format of Data Restored From Terminal Public Key Certificates

Sequence Number	Field Name	Length (in bytes)	Description	Format
1	Record Header	1	Hexadecimal '6A'	b
2	Certificate Format	1	Hexadecimal '07'	b
3	Acquirer Institution Identifier	4	IIN	n8
4	Merchant Identifier	15	MID, Merchant Identifier	an15
5	Terminal Identifier	8	TID, unique terminal identifier	an8

6	Certificate Expiry Date	2	Month and year (MMYY); certificate expires on last day of month	n4
7	Certificate Sequence Number	3	Unique identifier allocated by Root CA	N6
8	Hash Algorithm Identifier	1	Identifier of hash algorithm used to generate electronic signature value	b
9	Terminal Public Key Identifier	1	Terminal Public Key Identifier	b
10	Terminal Public Key Modulus Length	1	Terminal Public Key Modulus Length	b
11	Terminal Public Key Indicator Length	1	Terminal Public Key Indicator Length	b
12	Left-Side of Terminal Public Key Modulus	$N_{CA}-59$	If $N_{POS} \leq N_{CA}-59$, then this field contains a complete terminal public key modulus (N_{POS}) and the right side has been padded with a number of 'BB' bytes equal to $N_{CA}-59-N_{POS}$. If $N_{POS} > N_{CA}-59$, then this field contains the complete left-side section $N_{POS}-59$ of the terminal public key modulus;	b
13	Hash Value	20	Hash result of terminal public key and associated information; the hash result contains all of the information in this table in its listed order from the record header to the terminal public key indicator.	b

14	Restored Data End	1	Hexadecimal 'BC'	b
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- 2) Inspection of the Restored Data header; if it is not "6A" then card authentication of terminal legality fails.
- 3) Inspection of certificate format; if it is not "07", then card authentication of terminal legality fails.
- 4) Data elements 2 to 12 (from certificate format to left-side of terminal public key modulus) are connected from left to right; then, terminal public key remainder (if present) and terminal public key indicator are added to the end.
- 5) The specified hash algorithm (acquired from the hash algorithm indicator) is applied to the connected data described above, yielding a hash value.
- 6) The hash value that results from the above step is compared to the restored hash value; if they are not the same, then card authentication of terminal legality fails.
- 7) If the terminal public key algorithm identifier cannot be determined then card authentication of terminal legality fails.
- 8) If all of the inspections listed above are passed successfully, then connecting the left-side of the terminal public key and the terminal public key remainder (if present) will yield the terminal public key modulus.

8.5.2 Terminal Private Key Management

Terminal public/private key pairs are generated by the terminal; once a public key can be signed by CA then the public key certificate is downloaded into the terminal. The terminal must guarantee the security of the private key and protect against all kinds of attacks that may be conducted on it.

8.5.3 Signature Calculations

Signature calculation procedures are as follows:

- 1) During the GPO instruction special processing phase, the card uses the CA public key index received by the Send Record command to locate CA public key information, then uses the CA public key information to retrieve the terminal public key from the terminal public key certificate (a description of CA public key information during card personalization is available in Appendix A). If terminal public key restoration fails, then card authentication of terminal legality fails.
- 2) The card uses the terminal public key and associated algorithm to apply a specified restoration function (See Chapter 9, *UICS Basic Specification – Part 4* for an explanation of restoration functions) to the terminal signature received by the Send Record command. If the Restored Data end is not equal to "BC", then card authentication of terminal legality fails.

Table 4 Format of Data Restored From Signed Dynamic Data

Sequence Number	Field Name	Length (in bytes)	Description	Format
1	Restored Data Header	1	Hexadecimal '6A'	b
2	Signed Data Format	1	Hexadecimal '05'	b
3	Hash Algorithm Identifier	1	Identifies the hash algorithm used to generate the hash result	b
4	Transaction Date	3	Transaction Date (YYMMDD)	an
5	Card Random Number	8	The random number returned to the terminal from the card	b
6	Terminal Random Number	8	Random number generated by terminal	b
7	Pad Byte	N _{POS} -43	A number of pad bytes "BB" equal to (N _{POS} -43), where N _{POS} indicates terminal public key modulus length	b
8	Hash Value	20	Hash result of dynamic application data and associated information	b
9	Restored Data End	1	Hexadecimal 'BC'	b

- 3) Inspection of the restored data header; if it is not "6A" then card authentication of terminal legality fails.
- 4) Inspection of signature data format; if it is not "05" then card authentication of terminal legality fails.
- 5) The transaction data and transaction date retrieved by PDOL are compared; if they are not the same, then card authentication of terminal legality fails.

- 6) The card random number and the random number sent to the terminal are compared; if they are not the same, then card authentication of terminal legality fails.
- 7) Data elements 2 through 6 of the table above (from signature data format to terminal random number) are connected from left to right.
- 8) The specified hash algorithm (acquired from the hash algorithm indicator) is applied to the connected data described above, yielding a hash value.
- 9) The hash value that results from the above step is compared to the restored hash value; if they are not the same, then card authentication of terminal legality fails.

If all of the steps listed above are successful, then the card has successfully authenticated the legality of the terminal.

8.6 Newly Added Command Description

Send Record

——Command description:

The command is used during card verification of terminal legality;

——Usage Requirements and security:

This command can only be used after the Select Application command but before the GPO command; during all other phases the card must return the response code “6985”;

——Command Packet (Introduction to APDU Code and Parameters)

Table 5 Send Record Command Packet

Code	Value
CLA	‘00’
INS	‘DB’
P1	00: Send data command 80: Send last data command Other values: Retain Note: P2=00 mean effective, P2=01 means not effective
P2	00: Send certificate-associated data mode 01: Send signature data mode
LC	Variable length

DATA	- When P1=00 and P2=00, the card receives the signature algorithm identifier, terminal public key certificate and related information. For format details see Table 7; - When P1=00 and P2=80, the card receives the last section of relevant data and returns an 8-byte card random number. For format details see Table 7; - When P2=01, the card receives the terminal private key signature result and returns “9000”.
LE	None or 8

——Response Message Data

When P2=00 and P1=80, an 8-byte card random number is returned.

——Response Packet Status (SW1, 2)

Table 6 Send Record Response Packet Statuses

SW1	SW2	Meaning
90	00	Command executed correctly
6E	00	Incorrect CLA
6D	00	Incorrect INS
6A	86	Incorrect P1、P2 parameters
67	00	Incorrect length
69	85	Unsatisfactory security status
94	01	CA public key not found
94	02	Data field format incorrect

——Processing Procedures

- Once the card receives the Send Record command it will factor the data according to an L1L2L3 L4 L5 L6V1V2V3 V4 V5 V6 format (see Table 7); (Note: once an application has been selected, the first 6 bytes of the data field of the first Send Record command received by the card must be L1L2L3 L4 L5 L6, and mutual security authorization can only occur once each time an application is selected)

- 2) Once the card has received verified data it returns a random number to the terminal in response.
- 3) The terminal uses the terminal private key to sign the transaction date, card random number, and terminal random number; the Send Record command then transmits these signatures to the card;

—Data Fields

- 1) During the process of using the Send Record command, when P2=00 or 80, then command data fields will be organized according to a L₁ L₂ L₃ L₄ L₅ L₆ V₁ V₂ V₃ V₄ V₅ V₆ format; see Table 7.

Table 7 Data Format of Certificate-Related Data Send by Send Record Command

Data Do-main Field	Length (in bytes)	Description	Format
L ₁	1	Signature algorithm indicator length	b
L ₂	1	Transaction date length	b
L ₃	1	CA public key index length	b
L ₄	1	Terminal Public Key Indicator Length	b
L ₅	1	Terminal public key remainder length	b
L ₆	1	Terminal public key certificate length	b
V ₁	1	Signature algorithm identifier	b
V ₂	3	Transaction Date YYMMDD	an
V ₃	1	CA public key index	b
V ₄	1 or 3	Terminal public key index	b
V ₅	Variable	Terminal public key remainder	b
V ₆	Variable	Terminal public key certificate	b

It is best of the data in Table 7 is sent within a single command, but should the terminal public key certificate be too long then the terminal should cut the data domain into as few pieces as required to send with as few commands as possible.

Once the card receives the last Send Record instruction (P2=00, P1=80) then it will restructure and analyze the data.

- 2) When P2=01, the terminal will send the signature result as a data domain and the card will store it until the GPO instruction processing phase; the “Card Additional Processing” (Tag “9F68”) security mechanism will determine whether or not verification of the terminal signature result is necessary.

9 Sensitive Information Management

9.1 Overview

Enhanced low-value payment cards and terminals should all be equipped with necessary protective measures so as to safeguard sensitive electronic cash information. The bankcard industry must guarantee the security, integrity, and dependability of sensitive information and transaction data so as to guard against unauthorized usage, leakage, and data destruction. As contactless terminals become increasingly common it will become increasingly important for cards and terminals to protect the security of this kind of information.

9.2 Protection of Sensitive Transaction Data

Contactless devices and acquirer systems should be capable of storing all of the basic information necessary for low-value transaction sorting, especially as contactless devices cannot store or display personal IC electronic cash card information.

Contactless terminals are capable of reading the following electronic cash card data:

- PAN number (Tag5A)
- PAN sequence number (Tag5F34)
- Application expiry date (Tag5F24)
- Magnetic strip equivalent data (Tag57)
- Magnetic data (Tag9F1F)

Data that must be blocked includes:

- Last name (Tag5F20)
- Last Name Extension (Tag9F0B)
- Certificate number (Tag9F61)
- Certificate type (Tag9F62)

9.3 Card Security Requirements

It is recommended that blocked data elements not be written into enhanced low-value payment cards during the issuance process.

At the same time, during the personalization process contactless application file locators should not contain blocked data elements, this realizes blocking functionality for sensitive information.

9.4 Terminal Security Requirements

The complexity and type of the security mechanisms of contactless devices are determined by several different factors, such as device usage lifetime, device cost,

technology used (e.g. IC card functionality), acquirer system support of POS devices, etc. The following are the minimum security requirements of these types of devices:

- The display, storage, or printing of sensitive information is strictly prohibited;
- Terminal parameters must be updated;
- Source codes must be protected

Appendix A (Appendix of Specifications)

Card Personalization – Special Considerations

UnionPay standard IC card personalization content references *UnionPay Integrated Circuit Card Specifications—Auxiliary Specifications*, Section 1: Debit/Credit Application Personalization Templates and *UnionPay Integrated Circuit Card Specifications—Auxiliary Specifications*, Section 5: UnionPay IC Card Personalization Guide. Personalization process details will not be repeated here; this appendix only lists special considerations regarding enhanced low-value payment card personalization.

A.1 Special Personalization Considerations - Newly Added Data Elements

- DF61: Used to indicate a card's support of add-value functionality during personalization of applicable FCI information (a BFOC template) and return of an issuer's self-defined data BFOC. Terminals can use the Get Data command to acquire this data and can modify it using the issuer script command Put Data.
- Suggested value: 0x0C
- 9F69: Stored within a card's protected memory with a level of security equal to that of risk control parameters; used to store enhanced fDDA card authentication data and personalization using the same procedures as used for internal risk control parameters;
- When using enhanced fDDA, the first byte of this data element is "0x01", which indicates that the card supports enhanced fDDA; the second through fifth bytes should be "0x00000000"; the six and seventh byte is the Tag9F6C value; the eighth byte is the Tag DF61 value.

A.2 PDOL Data Setup – Special Considerations

- Contact channel PDOL setup: For currently supported contact electronic cash cards, the contact channel PDOL return value is 9F7A019F02065F2A02; 9F7A is the terminal electronic cash transaction support identifier; cards use this to determine whether or not a terminal supports electronic cash transactions; for UnionPay enhanced low-value payment products, due to the industry requirement of closed contact electronic cash transactions the contact channel PDOL value 9F7A01 has been discarded.
- Contactless channel PDOL setup: Aside from the PDOL requirements listed in *UICS Basic Specification – Part 5*, in order to guarantee the integrity of transaction dates it is required that terminals and cards support the following PDOL: "9F6604 9F0206 9F0306 9F1A02 9505 5F2A02 9A03 9C01 9F3704 9F2103 9F4E14"

Table 8 qUICS Most Basic PDOL Content

PDOL Tag	Data Element Name
'9F66'	Terminal transaction properties
'9F02'	Authorized amount
'9F03'	Other amount
'9F1A'	Terminal country code
'95'	Terminal Verification Result (TVR) Note: TVR will be filled with 0 by the qUICS reader (when requested data cannot be supplied by the reader it will still be processed in this manner)
'5F2A'	Transaction currency code
'9A'	Transaction Date
'9C'	Transaction type
'9F37'	Unpredictable number
'9F21'	Transaction time
'9F4E'	Merchant name